

GLACIS: FIELD GUIDE & PRACTICAL WORKFLOW REFERENCE

High-Speed Offline Security Assessment Worksheet · eJPTv2 / eCPPT / OSCP Practical Companion

What is GLACIS?

GLACIS is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate assessment cognitive overload. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax playbooks, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while GLACIS manages your operational memory.

INE & OffSec Exam Compliance

- **100% Passive & Offline:** Runs entirely on localhost via local SQLite database.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them.
- **Zero Cloud / AI Dependencies:** No external API calls, leaks, or prohibited LLMs.
- **Permitted Personal Notes:** Functions strictly as a local worksheet and syntax reference.

1. The Four Operational Stations (Switch with Keys 1, 2, 3, 4)

Station	Name & Purpose	What You See & Do Here	Hotkey
Station 1	Cockpit (Workbench)	Targets tree, open ports, service triage status, methodology roadmap, field notes, and interactive command console.	[1]
Station 2	Playbook Browser	Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc) with instant copy.	[2]
Station 3	Credential Matrix	2D grid of discovered credentials (rows) × target services (cols). Press [Enter] to compile spray commands; [Space] to cycle verification state.	[3]
Station 4	Proofs, Loot & Evidence	Track assessment question answers (:q), user/root flags, disk-backed loot browser ([Space]), screenshot proofing ([v]), and Failure Logs.	[4]

2. Station 1 Cockpit Architecture & Real-Time Workspace

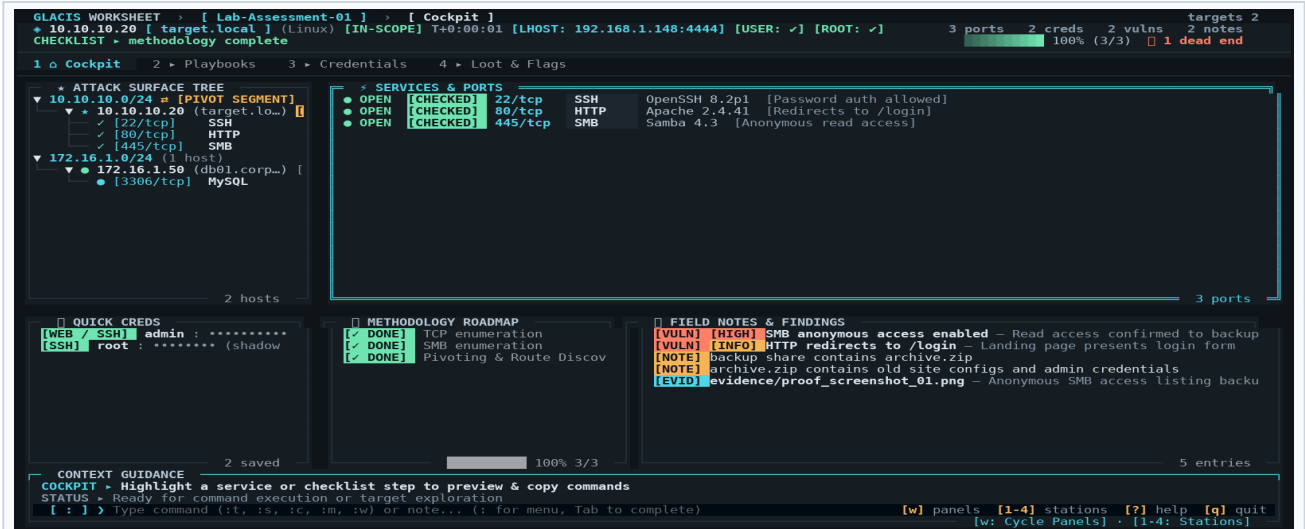


Figure 1: Station 1 Cockpit — Attack Surface Tree (left), Services & Port Triage (top), Methodology Roadmap (mid), and Bottom Command Console.

3. Keyboard Navigation Matrix ("Muscle Memory Map")

GLACIS is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless a text input is focused:

Key	Category	Action & Operational Behavior
1, 2, 3, 4	Station	Switch immediately to Station 1 (Cockpit), 2 (Playbooks), 3 (Creds Matrix), or 4 (Loot & Flags).
I (Shift+i)	Ingestion	Import Scan & Enum: Interactive review modal for Nmap (-oX, -oN, -oG) and Web Enum (FFUF, Ferox, Gobuster). Saves evidence to scans/ or enum/.
W (Shift+w)	Workspace	Workspace Manager: Switch between assessment workspaces or scaffold a new per-lab directory (scans/, enum/, screenshots/, loot/).
w	Navigation	Cycle Panels: Sequentially rotate focus across all panels in Cockpit without Tab.
h / l	Navigation	Column Jump: Press h for Sidebar (left); press l for Workbench (right).
j / k	Navigation	Vim Navigation: Move highlight down (j) or up (k) in the active list or tree.
[/]	Targeting	Target Switcher: Switch immediately to previous [or next] target machine.
b	Display	Sidebar Toggle: Collapse sidebar to give 100% width to Workbench (press b again to restore).
: (colon)	Console	Focus Command Bar: Instantly activates the bottom console input ready to type.
Esc	Console	Dismiss / Blur: Clears command input and returns focus directly to the active workbench.
Space	Triage / State	• On Service: Cycle triage status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED]. • In Station 3 Matrix: Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. • In Station 4 Loot: Open Loot Preview Modal to inspect hashes, configs, or keys.
, / .	Tool Carousel	Recipe Carousel: When a service is highlighted, press . (next) or , (prev) to cycle alternative tool recipes in the console (e.g. feroxbuster → gobuster → nikto → curl).
Enter	Execute / Copy	• On Service / Recipe: Copies the previewed tool command to clipboard. • In Station 3 Matrix: Compiles & copies ready-to-run spray command for credential & port. • In Station 4 Loot: Copies relative file path to clipboard (e.g. loot/users.txt).
v	Evidence	Paste Screenshot: In Station 4, saves clipboard image directly to screenshots/ and links as Evidence.
y	Quick Copy	Copies selected entity's primary value (IP address, port, password, or checklist command).
z	Layout	Zoom: Maximize the focused panel to full-screen; press z again to restore normal layout.
T (Shift+t)	Theme	Theme Picker: Open visual theme modal with 10 vibrant palettes (press d to set default).
/ or Ctrl+F	Search	Global Fuzzy Search: Search across targets, services, credentials, checklists, and notes.
r	Reference	Quick Reference Modal: Open offline practical reference for instant syntax lookup.
t / s / c / n	Fast Capture	Modal dialogs to quickly add Target (t), Service (s), Credential (c), or Note (n).
a	Question Proof	In Station 4: Open Add Proof modal to link question/item number to proof evidence.

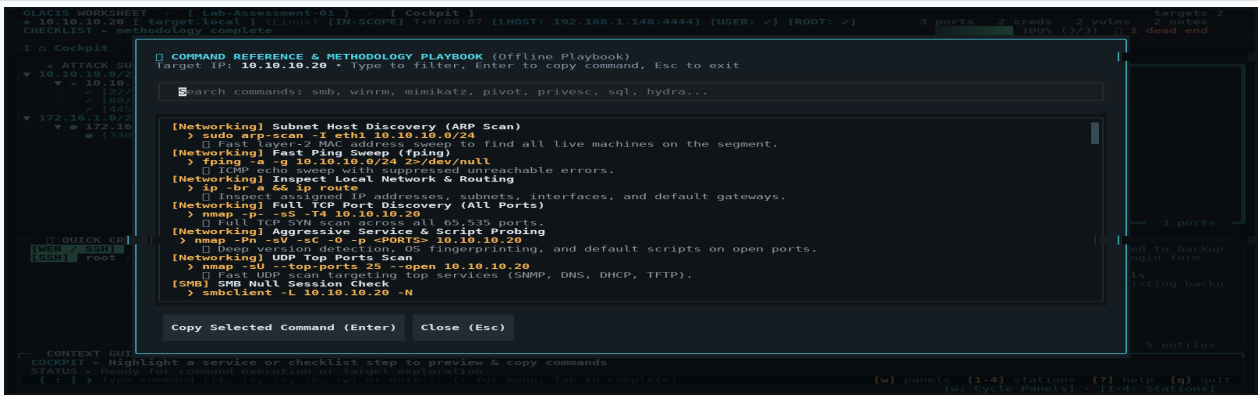


Figure 2: Practical Reference & Playbook Modal (Hotkey [r]) — Instant syntax lookup for Nmap, NetExec, Hydra, Chisel, and LinPEAS.

PRO-TIP: THE ZERO-MOUSE KEYBOARD FLOW

1. Highlight any port in **SERVICES & PORTS** with j / k.
2. Watch the bottom console display the exact syntax with the target IP already substituted.
3. Press . to cycle alternative tools (feroxbuster → gobuster → nikto → curl).
4. Press Enter to copy the command directly to your terminal clipboard, switch to your attack terminal, and run it!
5. Press Space to mark the port [CHECKED] or [DEAD-END].

4. Dual-Terminal Operating Model & Universal CLI Ingestion

The standard professional workflow runs GLACIS TUI in one terminal pane while executing active scans and exploits in an adjacent shell. Because GLACIS uses a shared SQLite store with multi-process notification, changes committed in your attack terminal update the TUI live.

Workflow & Goal	Command Line Execution (In Attack Terminal)	Operational Result in GLACIS
Automated Scan Ingest	<code>nmap -sC -sV \$IP -oX scans/box.xml && glacis import --apply scans/box.xml</code>	Parses all hosts, ports, services, and banners into SQLite. Raw XML archived into scans/ as immutable evidence.
1-Liner Shell Wrapper	<code>gnmap() { nmap -sC -sV "\$@" -oX scans/s.xml && glacis import --apply scans/s.xml; }</code>	Type gnmap 10.10.10.20: scans and populates GLACIS with zero manual typing.
Direct Terminal Pipe	<code>nikto -h \$IP glacis extract - --apply cat enum.log glacis extract - --apply</code>	Parses raw terminal text on the fly, extracting targets, ports, hashes, and CTF flags.
Target & Service Add	<code>glacis target 10.10.10.20 --os Linux glacis service 10.10.10.20 80/tcp http</code>	Direct CLI insertion from background scripts or terminal loops without opening the TUI.
Discovered Credential	<code>glacis cred admin:Password123! -t 10.10.10.20</code>	Immediately adds credentials into the Station 3 Credential Matrix.
Flag & Proof Capture	<code>glacis flag user 7f3a9e... -t 10.10.10.20 glacis flag root f04b12... -t 10.10.10.20</code>	Records objective flags with timestamps and target links in the Station 4 Loot Ledger.

Universal Static Ingest Formats (For Any Tool)

You are never locked to Nmap. Output from any tool (Masscan, Feroxbuster, Nikto, custom Python/Bash) can be converted to these standard formats and imported instantly:

Format Type	File Example / Syntax	How to Ingest & Import
Universal JSON import.json	<pre>[{"ip": "10.10.10.20", "hostname": "box.local", "os": "Linux", "services": [{"port": 80, "protocol": "tcp", "service": "http", "version": "Apache 2.4"}]}]</pre>	In TUI: <code>:import import.json</code> In CLI: <code>glacis import --apply import.json</code>
Universal Line Text import.txt	<pre># IP PORT/PROTO SERVICE VERSION 10.10.10.20 80/tcp http Apache 2.4 10.10.10.20 445/tcp smb Samba 4.3 10.10.10.30:3389:rdp:Windows RDP</pre>	Easily generated via <code>awk</code> or <code>sed</code> from any tool output. Ingest with <code>:import import.txt</code> .
Web Fuzzing JSON ffuf.json	<pre>ffuf -w wordlist.txt -u http://10.10.10.20/FUZZ -of json -o web.json</pre>	Auto-detects FFUF/Feroxbuster endpoints, response codes, and sizes in the review modal.

THE DEDICATED EVIDENCE ARCHITECTURE (scans/, enum/, loot/)

When you run `glacis init <box_name>`, GLACIS creates a dedicated folder with standard subdirectories: scans/ (Nmap, Masscan, Rustscan), enum/ (web fuzzing logs), screenshots/ (proof images), and loot/ (hashes, extracted creds). Every import command automatically archives raw tool outputs into these folders and computes SHA-256 hashes for exam submission integrity.

5. Bottom Console Syntax & Wordlist Accelerators

Console Command	Description & Operational Effect	Example Syntax
<code>:w <alias></code>	Wordlist Accelerator: Instantly copies standard Kali / SecLists paths to clipboard.	<code>:w rockyou :w common :w medium</code> <code>:w raft-d :w raft-f :w users</code>
<code>:import <path></code>	Scan & Enum Ingestion: Interactive review modal for Nmap XML/text or Web Enum. Archives raw output to scans/ or enum/.	<code>:import scans/nmap.xml</code> <code>:import enum/ffuf.json</code>
<code>:ws [name init]</code>	Workspace Manager: Switch active lab or scaffold standard folder layout.	<code>:ws lab01 :ws init lab02</code>
<code>:lhost / :lport</code>	Attacker IP & Port: Auto-detects VPN IP (tun0) and substitutes into all playbooks.	<code>:lhost auto :lport 4444</code>
<code>:t <ip> [opts]</code>	Add target with optional host, OS, subnet, and pivot flag.	<code>:t 10.10.10.5 host linux</code> <code>10.10.10.0/24 pivot</code>
<code>:s <port> <svc></code>	Record newly discovered port & service on active target.	<code>:s 445/tcp smb :s 8080 http</code>
<code>:c <user:pass></code>	Record discovered credential with optional service scope.	<code>:c admin:Secret123! SMB :c</code> <code>root:toor SSH</code>
<code>:c crack <id> <plain></code>	In-Place Cracking: Upgrades captured hash to plaintext secret without duplicating records.	<code>:c crack 1 Password123</code>
<code>:export wordlists</code>	Spray Wordlist Export: Generates deduplicated loot/users.txt and passwords.txt.	<code>:export wordlists :export creds</code>
<code>:paste-ev / :ev latest</code>	Screenshot Proofing: Saves clipboard image to screenshots/ or links newest OS capture.	<code>:paste-ev Root proof</code> <code>:ev latest Question 14</code>
<code>:q / :export</code>	Pin assessment question proof or export markdown evidence dossier.	<code>:q 7 /var/www/wp-config.php</code> <code>:export exam :export report</code>
<code>:uflag / :rflag <hash></code>	Record user flag or root/admin flag into Station 4 Loot Ledger.	<code>:uflag 7a9e... :rflag f04b...</code>
<code>:st <where> / :cl <clue></code>	Record stuck point (rabbit hole) and breakthrough clue.	<code>:st stuck on wp-login :cl found</code> <code>backup.zip</code>
<code>:m <template></code>	Instantiate standard checklist template (replace or append).	<code>:m ejpt :m web :m smb append</code>
<code>:theme <name></code>	Switch theme live (e.g. sugary, midnight, slate, caramel, cyber).	<code>:theme sugary :theme midnight</code>

6. Step-by-Step Practical Assessment Workflow ("The Battle Plan")

PHASE 1: SCOPING & HOST DISCOVERY

- Host Discovery:** Identify alive target IPs (nmap -sn 10.10.10.0/24 or arp-scan).
- Fast Add to GLACIS:** Type `:t 10.10.10.20` into console. Assign subnets: `:subnet 10.10.10.20 10.10.10.0/24`.
- Scope & Routing:** Tag dual-homed pivot machines with `:pivot 10.10.10.20 172.16.1.0/24` for automatic route guidance.
- Import Scan Option:** Bulk scans can be imported directly: `glacis import --apply scans/sweep.xml`.

PHASE 2: DETAILED PORT & SERVICE ENUMERATION

- Full Port Sweep:** `nmap -sS -p- --min-rate 1000 <IP>`, then version scan: `nmap -sV -sC -p <ports> <IP>`.
- Record Open Services:** Type `:s 80 http, :s 445 smb, :s 3306 mysql` into the console.
- Inspect Tactical Guidance:** Highlight each service in SERVICES & PORTS. Press `.` to cycle tool recipes (whatweb → feroxbuster → gobuster → nikto → curl). Press **[Enter]** to copy.
- Service Triage with [Space]:** Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].

PHASE 3: WEB & LOW-HANGING FRUIT EXPLOITATION

- Inspect Anonymous / Default Access:** Check anonymous FTP (`:s 21`), null SMB shares (`:s 445`), and SNMP strings (`:s 161`).
- Directory Fuzzing:** Type `:w common` or `:w medium` to copy the SecLists directory path. Paste directly into feroxbuster or gobuster.
- Cross-Filtering:** Highlighting a service in SERVICES & PORTS auto-scrolls the **Methodology** checklist to that service's testing steps!

PHASE 4: CREDENTIAL SPRAYING & LATERAL MOVEMENT (STATION 3)

- Record Every Credential:** Whenever you find creds (web configs, DB dumps, code), press **[c]** or type `:c user:pass scope`.
- Open Station 3 Matrix:** Press **[3]** to switch to the 2D Credential Spray Matrix.
- Instant Spray Generation:** Highlight any cell and press **[Enter]** to compile & copy ready-to-run spray commands (netexec, evil-winrm).
- Record Verification State:** Press **[Space]** on the cell to cycle: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. State persists in SQLite!

PHASE 5: PRIVESC, PIVOTING & EVIDENCE CAPTURE (STATION 4)

- Pivoting & Route Addition:** Tag pivot hosts with `:pivot` to see dynamic proxy guidance in the console bar.
- Privilege Escalation:** Switch to Station 2 (press **[2]**) and search Linux SUID, sudo -l, Cron jobs, or Windows tokens / services.
- Capture Question Evidence & Flags:** Log question proofs with `:q <num> <proof>` (or press **[a]** in Station 4). Record flags with `:uflag / :rflag`.
- Export Assessment Submission Bundle:** Run `glacis export -o report.md` or `:export exam` in console!

THE 20-MINUTE ROTATION RULE (EXAM GOLDEN RULE)

If you have been analyzing a single service for more than 20 minutes without a new lead: **STOP**. Press **[Space]** on the service to mark it [DEFERRED]. Type `:stuck <service description>` to log where you are stuck, press **[J]** to rotate to another machine or port, and harvest low-hanging fruit elsewhere. Most exam failures occur because candidates spend 3 hours trapped down an unexploitable rabbit hole!

7. Station 3 Deep Dive: 2D Credential Spray Matrix

In multi-machine assessments, credential reuse across SSH, SMB, WinRM, and Web portals is the primary lateral movement technique. Station 3 prevents credential amnesia by organizing all discovered credentials (rows) against all target services (columns) in an interactive 2D grid:



Figure 3: Station 3 Credential Matrix — Interactive 2D grid of discovered accounts against authenticating services with status badges.

Action / Key	Mechanism & Tactical Utility	Persistence & Assessment Value
Arrow Keys / Vim	Navigate freely across cells in the 2D grid.	Instant visual overview of which accounts were tested where.
[Space] (Cycle State)	Cycles through: [UNTESTED] → [VALID] (Green) → [PWN3D] (Blue) → [INVALID] (Plum).	Persisted instantly to local SQLite database. Never guess if you tried a password.
[Enter] (Copy Command)	Automatically compiles and copies the exact command line to your clipboard (e.g. netexec smb 10.10.10.20 -u admin -p 'P@ssword123').	Zero typos under stress; supports SSH, SMB, WinRM, RDP, MySQL, and MSSQL.
[c] (Fast Add)	Opens modal to record newly discovered credentials (or type :c user:pass [scope]).	Automatically expands the matrix with a new row across all existing targets.

LATERAL MOVEMENT WORKFLOW SUMMARY:

- 1. **Harvest:** Extract credentials from web app configs (wp-config.php, .env), DB dumps, or SAM registry hive.
- 2. **Input:** Press [c] in GLACIS to add the username and password (or run glacis cred user:pass in your attack terminal).
- 3. **Matrix Spray:** Press [3] to open the grid. Highlight the intersection of the new credential with port 445 (SMB) or 22 (SSH).
- 4. **Execute:** Press [Enter] to copy the pre-built NetExec/SSH command, paste and run in your terminal.
- 5. **Record:** If admin/pwn3d, press [Space] twice to set [PWN3D]. You now have your next pivot point!

8. Station 4 Deep Dive: Question Proofs & Loot Ledger

Practical exams require concrete evidence: hashes, specific directory names, database records, usernames, or service versions. Station 4 provides a dedicated ledger to link lab findings directly to question numbers, plus a Failure Log to manage stuck points:

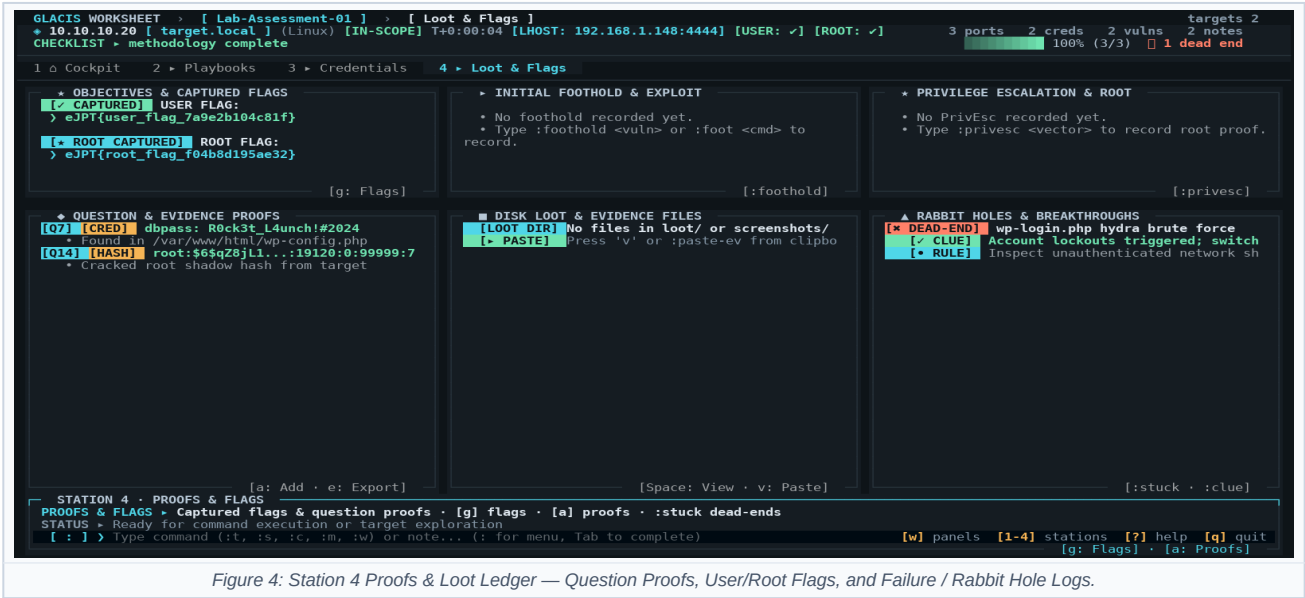


Figure 4: Station 4 Proofs & Loot Ledger — Question Proofs, User/Root Flags, and Failure / Rabbit Hole Logs.

Module / Feature	Hotkeys & CLI Syntax	Operational Purpose & Deliverable
Question Proofs	Press [a] (modal) or :q <num> <proof> [notes]	Links evidence directly to an exam question number (e.g. :q 14 /etc/passwd root hash). Prevents having to re-exploit targets at exam end.
Disk-Backed Loot Browser	Press [Space] to preview Press [Enter] to copy path	Scans loot/, screenshots/, enum/, and scans/. Preview text files, hashes, and dumps with instant copy.
Screenshot Proof Capture	Press [v] or :paste-ev [desc] / :ev latest	Saves clipboard image directly to screenshots/ via Wayland/X11 or auto-attaches most recent screenshot.
Loot & Flags Tracker	:uflag <hash> (user flag) :rflag <hash> (root flag)	Captures proof hashes with timestamps and associated target IP. Safely stored in local SQLite database.
Failure Log (Anti-Rabbit Hole)	:st <stuck_point> :cl <breakthrough_clue>	Records dead-ends and breakthroughs so you never repeat failed brute-force or injection attempts.
CLI Export Commands	glacis export -o report.md glacis export -f json -o bkp.json	Generates clean Markdown dossiers or complete JSON backups. Use --reveal-creds to include plaintext passwords.

THE OFFLINE SUBMISSION DOSSIER (glacis export -o report.md):
At any time, or before finalizing your exam answers in the INE / OffSec portal, run `glacis export -o report.md` in your terminal. GLACIS exports a complete, self-contained Markdown file containing: (1) Target Inventory & Subnet Map, (2) Solved Questions 1–35 with exact evidence strings, (3) Captured User & Root Flags, (4) Discovered Credentials & Verification States, and (5) Complete Audit Trail of all tested services. Keep this document open in your text editor while submitting exam questions for 100% verification confidence.

9. Station 2 Playbooks & Built-In Methodology Templates

Station 2 serves as an offline tactical encyclopedia. You can search playbooks for specific attack chains, inspect commands, and copy snippets with a single keystroke:



Figure 5: Station 2 Playbook Browser — Tactical guides, command recipes, and methodology references with instant keyboard copy.

Template Alias	Category	Key Checklist Steps & Focus
:m ejpt	eJPT Practical	Scope recon → Host discovery → Full TCP scan → Service triage → Low-hanging fruit → Web fuzzing → Cred spray → Pivoting → PrivEsc.
:m web	OWASP Web App	Technology profiling (whatweb) → Directory fuzzing (feroxbuster) → Parameter fuzzing → SQLi → Auth bypass → LFI/RFI → File upload.
:m smb	SMB & Windows	Null session shares (smbclient/smbmap) → User enum (rpcclient) → Password policy → Anonymous signing audit → Vulnerability scan (MS17-010).
:m pivoting	Network Pivoting	Dual-homed adapter discovery (ip a) → Internal route inspect → Proxychains & Chisel server setup → Port forward (socat) → Subnet sweep.
:m privesc_linux	Linux PrivEsc	sudo -l commands → SUID/SGID binaries (GTFOBins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports.
:m privesc_windows	Windows PrivEsc	whoami /priv (Selpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey) → Registry backup.

EXAM / ASSESSMENT START PROCEDURES:

1. **Launch GLACIS:** Open terminal and launch glacis (or short alias gls).
2. **Scaffold Dedicated Workspace:** Run glacis init target_box to create a fresh directory with scans/, enum/, screenshots/, and loot/.
3. **Theme Selection:** Press [T], select palette (e.g. Sugary, Midnight, Slate, or Cyber), press [d] to save as default.
4. **Dual-Terminal Ingestion:** Run scans in your attack terminal and auto-import: glacis import --apply scans/sweep.xml.
5. **Offline Markdown Dossier:** Run glacis export -o report.md or :export exam in the console at any time to verify all answers.
6. **Database Safety:** Stored locally in SQLite. 100% local, persistent, zero cloud calls, and fully compliant with exam rules.